

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **distributed network modelling** components in Java, focusing on high-throughput data processing for large-scale systems.
- Wrote **C++ code** for performance-critical backend services, contributing to the development of large-scale infrastructure and distributed systems.
- Participated in **design reviews** with cross-functional teams to evaluate and select appropriate technologies for complex system architectures.
- Reviewed code developed by peers, providing constructive feedback on style guidelines, accuracy, and testability to maintain high code quality.
- Analyzed and resolved **product or system issues** by debugging and tracing root causes impacting hardware, network, or service operations.
- Contributed to the creation and refinement of **technical documentation** and educational content based on product updates and user feedback.
- Containerized distributed system components using **Docker** and managed deployments on **AWS EC2** instances, ensuring efficient resource utilization.

SKILLS

Languages – C++, Java, Python, SQL, R

Distributed Systems – Large-scale infrastructure, Distributed computing, Network modelling, System design, Compute technologies, Storage architecture

Development Practices – Software development, Algorithms, Data structures, Code reviews, Agile methodologies, Debugging, CI/CD, Git

Cloud & Containerization – AWS (EC2, S3, Lambda, CloudWatch), Docker

Databases – PostgreSQL, MySQL, Oracle

Tools – Jenkins, Jira, Confluence, Splunk, Dynatrace

PROJECTS

Distributed Network Simulation Platform

- Developed a **Java-based simulation engine** to model network traffic patterns for large-scale distributed systems, processing over 10,000 simulated events per second.
- Implemented **data structures and algorithms** for efficient packet routing and congestion analysis within the simulation environment.
- Integrated **Docker** for containerizing simulation nodes, enabling scalable and reproducible testing across different network configurations.

Scalable Compute Resource Orchestration

- Engineered a **Python and Java service** to orchestrate compute resources on **AWS EC2** for distributed task processing.
- Designed RESTful APIs using **Spring Boot** for managing task submission, status tracking, and result retrieval, serving an internal team of 8 engineers.
- Wrote **shell scripts** for automating deployment and configuration of compute nodes, reducing manual setup time by 40%.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)